

NOTRE DAME OF MARYLAND UNIVERSITY

Health Sciences Curriculum Gap Analysis

A benchmarking brief comparing NDMU's undergraduate Health Sciences curriculum with recent U.S. health science program trends and Maryland health priorities.

FOCUS

NDMU BS Health Sciences curriculum and program positioning

BENCHMARK

Recent U.S. undergraduate health science curriculum scan

OUTPUT

Strengths, gaps, Maryland fit, and action roadmap

Key Judgments

NDMU's Health Sciences program has a strong interdisciplinary foundation, a useful liberal arts orientation, and an applied community-facing capstone. The main opportunity is to make its Maryland-facing value proposition more explicit through pathways, data literacy, field experience, and alignment with state health priorities.

- 01 NDMU is well positioned, but not yet visibly differentiated.** The program already combines sciences, communication, ethics, psychology, cultural humility, and healthcare systems content. Peer programs make their distinctive routes more visible through tracks, concentrations, internships, and microcredentials.
 - 02 The capstone is a real strength.** HSC-461 asks students to review literature, interview community members, and develop, implement, and evaluate a health promotion project. That aligns well with current applied-learning trends.
 - 03 Pathway architecture is the biggest gap.** Advisor-guided electives are flexible, but students and external stakeholders need clearer maps into pre-clinical graduate study, community health, behavioral health, health systems, and data-informed care roles.
 - 04 Data, digital health, and population health analytics need more visibility.** National peers increasingly foreground statistics, epidemiology, informatics, AI/data interpretation, dashboards, and health technology.
 - 05 Maryland gives the program a strong strategic anchor.** The state's current SHIP priorities are chronic disease, access to care, women's health, violence, and behavioral health. These can become capstone themes, internship targets, and pathway labels.
 - 06 The recommended identity is Maryland Health Equity and Community Health Systems.** This positioning fits NDMU's Baltimore location, liberal arts mission, ethics orientation, and the region's health system ecosystem.
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NDMU Curriculum Snapshot

41-43

Major credit hours listed for the BS Health Sciences program

29

Core Health Science credits across nine required courses

2023

Program launch year reported by NDMU news

5

Maryland SHIP priority areas available for curriculum alignment

What NDMU Requires

The major combines anatomy and physiology, introduction to health science, health science communications, legal/current trends, research in health care assessment/planning/evaluation, psychology, and a capstone.

Students also select one science elective and three interdisciplinary electives from areas such as microbiology, teamwork and negotiation, the U.S. health care system, cultural humility, topics in health sciences, and sociology.

What the Program Signals

NDMU's catalog frames the major as interdisciplinary, housed in biology, and aligned with critical thinking, diversity and identity, civic responsibilities, and preparation for employment or graduate study.

Program materials point to roles such as patient care advocate, health services manager, health communicator, health educator, clinical researcher, and public health advocate.

Interpretation: the curriculum has a coherent base, but the public-facing structure still reads as a broad exploratory major rather than a sharply positioned Maryland health workforce program.

How NDMU Compares with Peer Trends

Curriculum Dimension	Recent Peer Trend	NDMU Current Position	Implication
Visible pathways	Programs increasingly offer tracks such as pre-clinical, public/community health, rehabilitation, health administration, or health technology.	Electives are advisor-guided, but the catalog does not present formal tracks or pathway maps.	High-priority gap: create pathway sheets before adding many new courses.
Applied learning	Internship, co-op, fieldwork, service learning, capstone, and portfolio experiences are credibility markers.	Capstone is applied and community-facing, but a separate internship/fieldwork requirement is not clearly visible.	Good base: add an optional or restricted-elective internship tied to Maryland partners.
Capstone/research	Strong programs scaffold research methods, statistics, communication, and capstone artifacts.	HSC-373 and HSC-461 provide a useful research-to-capstone sequence.	Strength: preserve and brand as a Maryland Health Priorities Capstone.
Data and digital health	Statistics, epidemiology, informatics, digital health, and AI/data literacy are growing differentiators.	Basic statistics appears in general education support; digital health and informatics are not prominent in requirements.	Moderate-to-high gap: add a visible data/digital health module or course.
Health equity and systems	Programs increasingly require health systems, policy, ethics, equity, and population health content.	NDMU has Medical Ethics, Cultural Humility, U.S. Health Care System, and Legal/Current Trends options.	Strength with upside: make equity and systems alignment explicit in outcomes.
Microcredentials	Some programs use short credentials such as EMT, Mental Health First Aid, CPR/BLS, or portfolio badges.	No prominent microcredential stack is visible from catalog evidence.	Emerging opportunity: add low-credit or co-curricular credentials without major governance burden.

Peer trend baseline draws from prior official-source scan of eight U.S. undergraduate health science programs.

What to Preserve and What to Address

Strengths to Preserve

- Interdisciplinary structure housed in biology, with liberal arts breadth.
- Strong fit with ethics, communication, diversity, civic responsibility, and cultural humility.
- Applied capstone that includes literature review, community interviews, implementation, evaluation, presentations, and written synthesis.
- Clear preparation for both workforce entry and graduate health professions pathways.
- Baltimore/Washington location with access to hospitals, public health agencies, community organizations, and health nonprofits.

Gaps to Address

- No formal track structure visible to students, families, employers, or graduate programs.
- No clearly required internship, practicum, or fieldwork experience beyond the capstone.
- Limited visible emphasis on health informatics, AI/data interpretation, epidemiology, population health dashboards, or digital health ethics.
- Maryland SHIP priorities are not yet mapped directly to course outcomes, capstone themes, or partner experiences.
- No visible stack of microcredentials or portfolio artifacts that students can show before graduation.

The practical goal is not to replace NDMU's current curriculum. It is to convert its existing strengths into a clearer external promise and add a few high-signal elements that peers already use.

A Stronger Maryland-Facing Identity

Recommended program identity: **Maryland Health Equity and Community Health Systems**. This gives the program a distinctive regional purpose while preserving pre-health flexibility.

Why This Fit Is Strong

Maryland's 2024-2029 State Health Improvement Plan identifies five priority areas: chronic disease, access to care, women's health, violence, and behavioral health. NDMU can use these priorities as the organizing logic for capstone topics, internship sites, community partnerships, and student portfolio themes.

Baltimore and the Washington region also give students access to a dense ecosystem of hospitals, public health agencies, community organizations, research institutions, and health policy employers. NDMU's existing emphasis on ethics, communication, diversity, and civic responsibility fits that ecosystem better than a generic pre-health major would.

Signature Differentiators

Health equity	Community health systems	Behavioral health	Access to care	Women's health	Chronic disease
Violence prevention	Data literacy	Care coordination			

Academic Message

Students learn health science through biology, behavioral science, ethics, communication, research, systems thinking, and community-engaged practice.

External Message

Graduates are prepared to navigate Maryland's healthcare and public health ecosystem as ethically grounded, data-aware, community-facing health professionals.

Recommended Curriculum Architecture

The most efficient redesign is a pathway overlay first, followed by targeted curricular additions. This avoids unnecessary disruption while making the program easier to explain and advise.

Proposed Pathway	Student Audience	Current Courses to Anchor	High-Value Additions
Pre-Clinical and Graduate Health Professions	Students aiming for PA, OT, PT, nursing, pharmacy, medicine, public health, or related graduate programs.	BIO-201/202, HSC-150, HSC-373, HSC-461, genetics, microbiology, psychology.	Prerequisite planning sheets, graduate-readiness portfolio, clinical exposure or shadowing reflection.
Community Health, Behavioral Health and Health Equity	Students interested in community health, patient advocacy, behavioral health, public health, nonprofits, and health education.	HSC-201, PHL-339, LSP-150, SOC-101, HSC-310, HSC-461.	Maryland SHIP capstone themes, Mental Health First Aid, CHW introduction, community partner internship.
Health Systems, Data and Care Coordination	Students interested in healthcare management, care navigation, quality improvement, clinical research, population health, or health technology.	BUS-380, BUS-334, MAT-215, HSC-320, HSC-373, COM-106.	Health informatics/data literacy module, population health dashboard assignment, privacy/AI ethics case.

The pathway overlay can be piloted as advising maps before becoming formal concentrations. This creates immediate clarity while preserving faculty flexibility.

Action Roadmap

0-6 Months

- Create a one-page curriculum map linking current courses to Maryland SHIP priorities.
- Publish three advising pathway sheets using existing requirements and recommended electives.
- Rename or frame HSC-461 as a Maryland Health Priorities Capstone in advising materials.
- Add one health data literacy assignment to HSC-373 or HSC-320.

6-18 Months

- Build an 80-100 hour internship or field experience option with hospitals, local health departments, nonprofits, or community organizations.
- Add a required or restricted-elective module on health informatics, dashboards, AI/data interpretation, privacy, and digital health ethics.
- Bundle Mental Health First Aid, CPR/BLS, CHW introduction, or Spanish for Health Professionals into a visible skills portfolio.

18-36 Months

- Convert advising pathways into formal tracks if enrollment and faculty capacity support it.
- Create a partner advisory group from Maryland healthcare, public health, behavioral health, and community organizations.
- Assess student artifacts with a competency map covering evidence, communication, equity, systems, and applied practice.

First priority: build the curriculum map. It will reveal what is already strong, what is missing, and which changes require faculty governance versus advising or co-curricular action.

Caveats

- This brief is a curriculum benchmarking and positioning analysis, not an accreditation review.
- NDMU catalog and program pages were prioritized for current requirements; news and ecosystem sources were used for context.
- The peer trend scan is a limited benchmark sample, so findings should be read as planning signals rather than national generalizations.
- Credit rules, prerequisites, catalog copy, and Maryland policy priorities should be rechecked before formal proposal drafting.

NDMU and Maryland Sources

NDMU Catalog: Health Sciences Major

catalog.ndm.edu/health-science/major/health-sciences

NDMU Course Page: HSC-461 Capstone

catalog.ndm.edu/biology/hsc-461

Maryland Department of Health: Building a Healthier Maryland

health.maryland.gov/pha/Pages/Building-a-Healthier-Maryland.aspx

NDMU Program Page: Health Sciences

ndm.edu/academics/school-arts-sciences-business/health-sciences

NDMU News: Health Sciences Degree Launch

ndm.edu/news-and-events/news/new-health-sciences-degree-prepares-ndmu-students-variety-healthcare

Johns Hopkins Community Impact

communityimpact.jhu.edu

Peer Curriculum Sources

University of Florida Catalog

catalog.ufl.edu/UGRD/colleges-schools/UGPBH/HES_BHS/

Drexel University Catalog

catalog.drexel.edu/undergraduate/collegeofnursingandhealthprofessions/healthsciences/

Ohio State University

hrs.osu.edu/academics/undergraduate-programs/health-sciences/curriculum

DePaul University Catalog

catalog.depaul.edu/programs/health-sciences-bs/

University of Minnesota Rochester

r.umn.edu/academics/undergraduate-programs/bachelor-science-health-sciences

Stockton University

stockton.edu/health-sciences/health-science.html

University of Central Florida

healthprofessions.ucf.edu/healthsciences/bachelors/

Quinnipiac University

qu.edu/schools/health-sciences/programs/bachelors-degree/health-science-studies/

Design references: Stanford HAI 2025 AI Index Report, hai.stanford.edu/ai-index/2025-ai-index-report; MIT Work of the Future Report, workofthefuture-taskforce.mit.edu/research-post/the-work-of-the-future-building-better-jobs-in-an-age-of-intelligent-machines/; Harvard Kennedy School Belfer Center Report Page, belfercenter.org/research-analysis/eye-storm-pakistans-balancing-act-evolving-geopolitical-order.